

Subject: Dataset Expansion Update — Panvel NGO Network & Whisper STT Pipeline

We're scaling our audio data collection infrastructure across Panvel's special education network.

Current Outreach Status:

- Primary Anchor NGO: [Jai Vakeel Foundation] — where we conducted our Major project fieldwork and established our baseline protocol
- 2 Additional NGOs now in active contact for data partnership
- 3-site network forming a triangular coverage zone across Panvel

The Numbers:

- 200+ students per NGO → 600+ total potential participants across the network
- That's a 3:1 expansion ratio from our original single-site dataset
- With an estimated 60:40 split of intellectually disabled students across mild-to-moderate vs. severe categories, we're targeting roughly 360+ relevant audio profiles for our training corpus
- If we capture just 15 minutes of structured speech per child, we're looking at 90+ hours of raw audio — a 15x multiplier on our current sample size

Technical Context:

We're running OpenAI's Whisper architecture for our STT (Speech-to-Text) pipeline, specifically tuning for atypical speech patterns common across the intellectual disability spectrum. The model's robust encoder gives us a strong baseline, but the diversity of this 3-site, 600+ student corpus is what will let us fine-tune for the phonetic variability and prosodic differences we observed in our Major project fieldwork.

Next Step: Coordinating consent protocols and recording environment standardization across all three NGOs to maintain a 1:1 data quality ratio with our original site.

